AYE
TECH HUB

• CRITICAL THINKING SERIES • NO. 01

Introduction to Critical Thinking in the AI Age

Think Deeper. Question Everything. Stay Human.

No. 01 of 30 | Critical Thinking Series

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- Why critical thinking is the most valuable skill in 2026
- The 6 core critical thinking skills — and how to practise them
- How AI amplifies both good thinking and bad thinking
- The Paul-Elder framework — a universal thinking tool
- Your 10-day critical thinking starter challenge

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Why Critical Thinking Has Never Mattered More

We live in the most information-rich era in human history. Every day we are exposed to more data, more opinions, more headlines, and more AI-generated content than any previous generation ever encountered in a lifetime. And yet — the ability to **think clearly, question evidence, and reach sound conclusions** is declining, not improving.

AI tools like ChatGPT, Gemini and Claude can write essays, generate research, produce code, and answer almost any question in seconds. This is extraordinary. But there is a danger hidden in that convenience: when we outsource our thinking to machines, **we risk losing the very capacity that makes us human** — the ability to reason, doubt, create, and judge for ourselves.

> ***"The function of education is to teach one to think intensively and to think critically. Intelligence plus character — that is the goal of true education."***
>
> — Martin Luther King Jr.

What Is Critical Thinking?

Critical thinking is **the disciplined process of actively and skilfully analysing, evaluating, and synthesising information** gathered from observation, experience, reflection, or communication — as a guide to belief and action. It is not about being negative or argumentative. It is about thinking *well*.

Figure 1 — The 6 Core Critical Thinking Skills (Facione, 1990)

Skill	What It Means	Example in Practice
Analysis	Breaking a complex idea into its component parts to understand its structure	Identifying the claims, evidence and assumptions in a news article
Evaluation	Assessing the credibility of sources and the strength of arguments	Asking: who wrote this, what is their evidence, do they have a bias?
Inference	Drawing reasonable conclusions from available evidence	Concluding that a study is unreliable because its sample size is 12 people
Explanation	Clearly communicating your reasoning so others can assess it	Writing a structured argument that shows your evidence and logic
Self-Regulation	Reflecting on your own thinking process and correcting errors	Noticing you agreed with an article only because it confirmed your existing view
Interpretation	Understanding the true meaning and significance of information	Reading between the lines of a political speech to find unstated assumptions

Important: Critical thinking is a skill, not a talent. It can be learned, practised, and improved. Like physical fitness, it requires regular exercise. The remaining 29 issues in this series will give you exactly that exercise.

AI and Human Thinking — Amplifier, Not Replacement

AI does not replace your thinking — it **amplifies it**. If you are a careful, sceptical, evidence-based thinker, AI will make you dramatically more productive and better informed. If you are a lazy, credulous, confirmation-bias-driven thinker, AI will make you dramatically more wrong, more confidently.

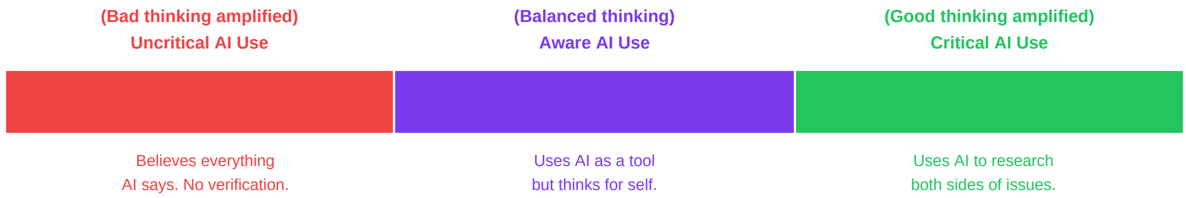

Figure 2 — How AI Amplifies Your Thinking Style

AI Behaviour	Uncritical User Thinks	Critical Thinker Knows
AI gives a confident, detailed answer	"It must be correct — it sounds authoritative."	"Confident tone means nothing. I need to verify the source and facts."
AI writes a persuasive essay on a topic	"This essay is great. I will submit it."	"This essay was generated statistically. It may contain errors, biases or fabricated citations."
AI says "studies show that..."	"OK, science confirms it."	"Which studies? What sample size? Who funded them? Is this peer-reviewed?"
AI produces two different answers to the same question	"The second one must be better — it is longer."	"AI is inconsistent. I need to investigate which answer is actually supported by evidence."
AI agrees with everything you say	"Finally, something that understands me."	"AI is trained to be agreeable. Agreement is not validation. I need a real challenge to my view."

The Paul-Elder Framework — A Universal Thinking Tool

Developed by Richard Paul and Linda Elder at the Foundation for Critical Thinking, this framework gives you 8 questions to ask about **any** piece of reasoning — whether from a person, a newspaper, an AI chatbot, a government, or your own mind.

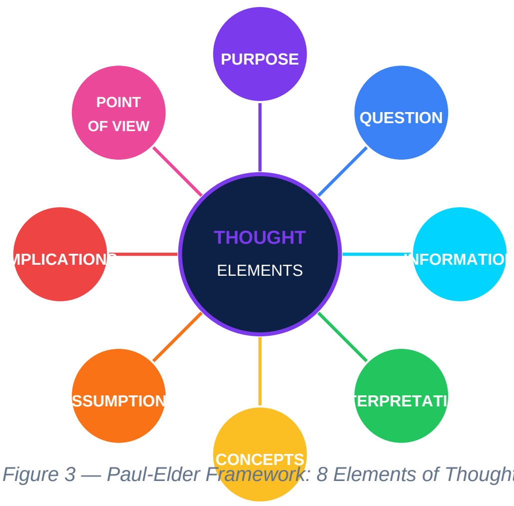

Figure 3 — Paul-Elder Framework: 8 Elements of Thought

How to use it: Next time you read a news article or receive an AI-generated response, work through all 8 elements. Ask: What is the PURPOSE of this content? What QUESTION is it answering? What INFORMATION is it using — and what is it leaving out? What ASSUMPTIONS is it making? What are the IMPLICATIONS if this is true?

The Enemies of Clear Thinking

Before you can think better, you must understand what is currently preventing you from thinking clearly. These are not character flaws — they are **features of the human brain** that evolved for a world very different from the one we live in today.

Enemy	What It Is	AI Age Example
Confirmation Bias	The tendency to search for, interpret and remember information that confirms what you already believe	Your AI search assistant learns your views and only shows you content that agrees with them
Availability Heuristic	Judging how likely something is based on how easily an example comes to mind	After reading AI-generated scary headlines, you overestimate real-world dangers
Authority Bias	Accepting claims because they come from a perceived expert or authoritative source	Trusting AI output because it sounds confident and professional
Dunning-Kruger Effect	People with limited knowledge overestimate their competence; experts underestimate theirs	Using AI to write on a topic you know little about then presenting it as expertise
Groupthink	The desire for harmony in a group leads to irrational or poor decision-making	Social media algorithms create echo chambers where all your "community" agrees
Sunk Cost Fallacy	Continuing a course of action because of past investment, not future value	Defending an AI-generated wrong answer because you spent time on it

Your 10-Day Critical Thinking Starter Challenge

Day	Challenge	Purpose
1	Find one news story and identify: Who wrote it? What evidence is used? What is missing?	Practise source evaluation and gap identification
2	Ask an AI a question, then verify every fact it gives you independently.	Build the habit of not accepting AI output at face value
3	Find one belief you hold strongly. Write down 3 good arguments AGAINST it.	Counter confirmation bias by steelmanning opposing views
4	Watch a debate or interview. List the logical fallacies you hear.	Develop awareness of flawed reasoning in real arguments
5	Spend 30 minutes reading a viewpoint you strongly disagree with — charitably.	Expand your intellectual empathy and reduce tribalism
6	Apply all 8 Paul-Elder elements to one article or AI response.	Practise the universal thinking framework
7	Have a conversation where your only goal is to understand, not to win.	Develop listening skills and reduce ego in debate
8	Write down 5 assumptions you are making about your own life or career.	Develop self-aware thinking and reduce blind spots
9	Take one common "fact" you have never questioned. Research its actual evidence.	Challenge assumed knowledge

10	Reflect: What did you believe 5 years ago that you now know is wrong?	Appreciate intellectual growth and stay humble about current beliefs

What comes next — CT-002: Cognitive Biases: The Errors Your Brain Makes Daily — a deep dive into the 20 most dangerous cognitive biases, with real-world examples and practical strategies to overcome each one.